ECON 0100 | Vignette E1 | Costs

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

As an economics student at Hogwarts, the Ministry of Magic has tasked you with analyzing the impact of policy proposals on three firms in the cauldron industry, given their distinct cost structures:

The Cauldron Co. is a large, established firm producing high-quality cauldrons with traditional methods. They have high fixed costs but low, constant marginal costs due to economies of scale.

The Witching Hour is a smaller firm specializing in novelty cauldrons. They have moderate fixed costs but increasing marginal costs due to specialized production.

Nimbus Cauldron Works is a new entrant producing low-cost, mass-produced cauldrons made to order by a contract foundry. They have no meaningful fixed costs, and low constant marginal costs.

Q1 | The Witching Hour's Workshop

The table shows how many cauldrons The Witching Hour makes in a day with each number of workers. Each worker is paid 60 coins a day, and the workshop pays 120 coins a day in rent.

L	Q	Marginal Product	Marginal Cost	Variable Cost	Total Cost	ATC
0	0	—	—			—
1	15					
2	24					
3	30					
4	32					

Fill in the marginal product of each worker. Then fill in the marginal cost, variable cost, total cost, and average total cost at each number of workers.

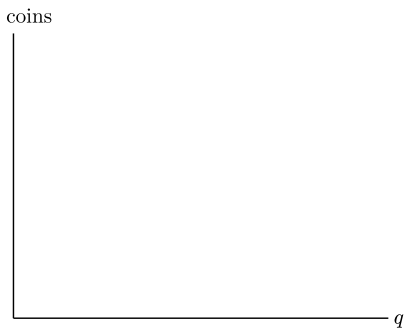
a) What happens to the marginal product as workers are added?

b) Why does The Witching Hour have increasing marginal costs?

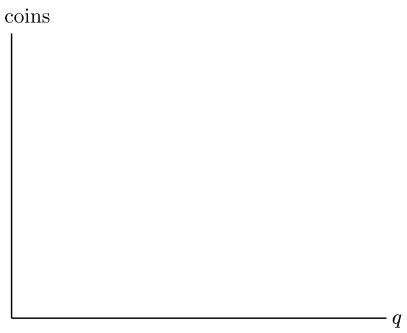
Q2 | Cost Curves and Policies

Graph all three firms' average total cost and marginal cost curves. Then show how each firm's curves shift with the policy listed.

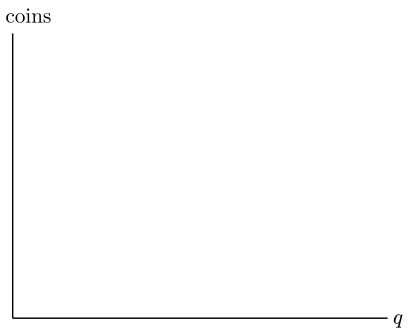
The Cauldron Co.
(lump sum tax)



The Witching Hour
(per unit tax)



Nimbus Cauldron Works
(per unit subsidy)



ECON 0100 | Vignette E2 | Competition

Q1 | A Higher Marginal Cost

The Witching Hour has a marginal cost of $MC = 10 + 2q$. Many identical workshops sell identical cauldrons, so each takes the market price as given.

a) At $P = 50$ what quantity maximizes the firm's profit?

b) Calculate the firm's maximal profit if $ATC = 50$ at the profit-maximizing quantity.





Now suppose the Ministry taxes every cauldron 10 coins, so marginal cost rises: $MC = 20 + 2q$. Start with identical firms earning zero economic profit in long-run equilibrium.

c) What happens to price, each firm's output, and profit in the short run after the new cost?

d) In the long run the price rises to 60. What quantity maximizes each remaining firm's profit?

e) What happens in the long run to price, each the firm's output, and market output?